



Developed By- Michigan Department of Health & Human Services

Advanced Molecular Detection Southeast Region

This resource was made possible through funding provided under the Epidemiology and Laboratory Capacity for Prevention and Control of Emerging Infectious Diseases (ELC) Cooperative Agreement (CK24-0002), Project D: Advanced Molecular Detection to the Florida Department of Health. The conclusions, findings, and opinions expressed by authors do not necessarily reflect the official position of the U.S. Department of Health and Human Services, the Public Health Service, or the Centers for Disease Control and Prevention.

UPDATES

2026 Molecular Epidemiology Training Series

- 2026 Training Series hosted by the Southeast Region BRR & University of Georgia
- First session held on August 21, 2026
- Series runs August 21–December 4 with 5 modules (lecture, case study, practical)
- Training materials available on GitHub: <https://github.com/StaPH-B/southeast-region/tree/master/trainings/2026%20ELC%20Training>
- Registration link: <https://forms.gle/h4g1uV8PkWqDMrme8>
- Reach out with any questions.

OVERVIEW

Purpose:

BaSSeT, developed by Michigan Department of Health & Human Services, provides automated bacterial genome analysis, including species identification, serotyping, virulence/AMR profiling, and organism-specific typing.

Usage:

Used by public health labs and researchers to process assemblies or sequencing data for surveillance, outbreak investigations, and routine genomic typing.

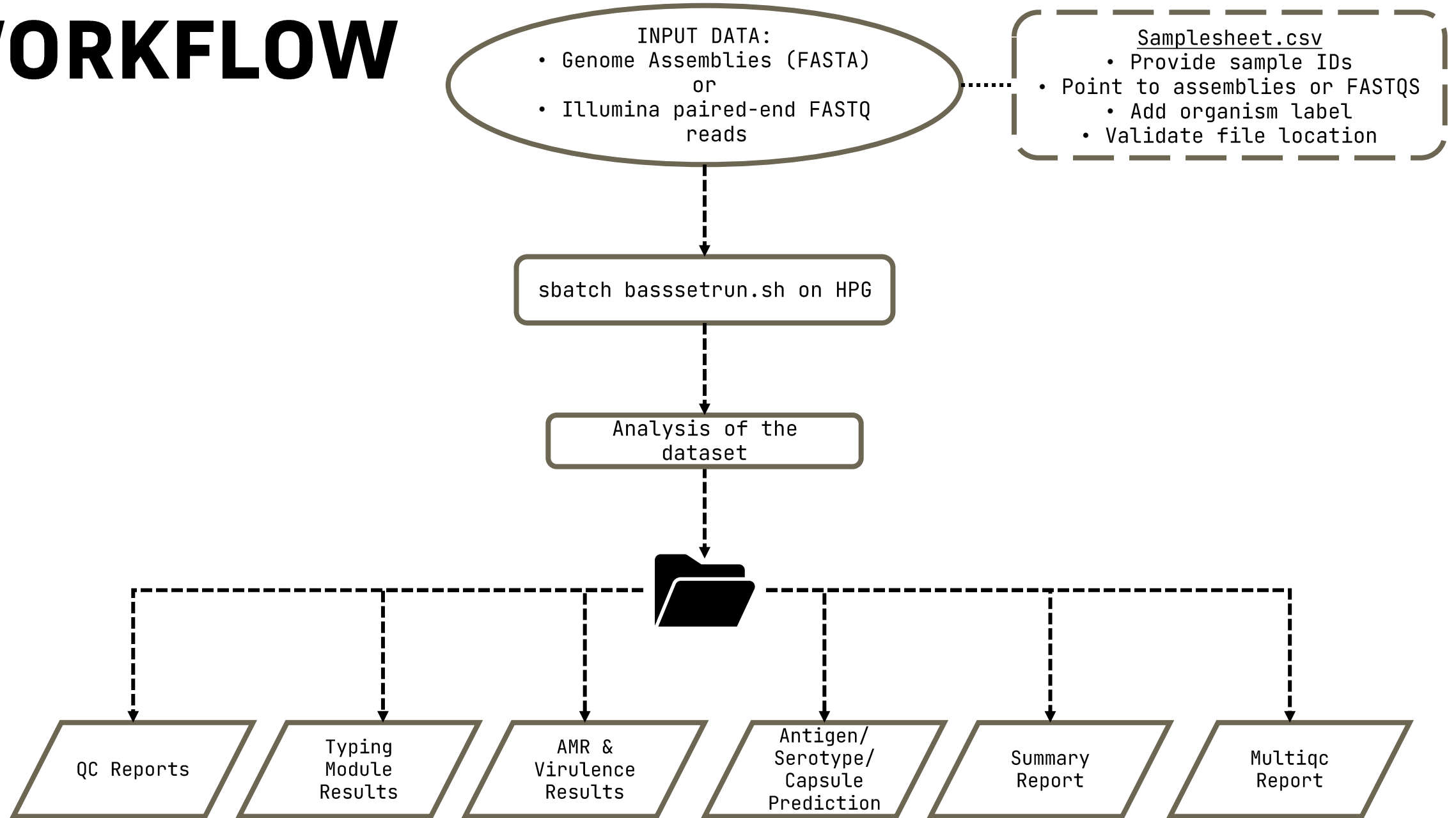
Dependencies:

Nextflow

Conda

Apptainer/Singularity

WORKFLOW



Application

OBJECTIVE

Use various genome assemblies (FASTA) and analyze the dataset using BaSSet pipeline

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/

Name	Size	Changed
..		7/21/2026 12:04:46 PM
assets		7/17/2026 6:14:25 AM
bin		7/17/2026 6:14:25 AM
cache		7/22/2026 12:19:45 PM
conf		7/17/2026 6:14:25 AM
docs		7/17/2026 6:14:25 AM
input		7/22/2026 12:04:39 PM
lib		7/17/2026 6:14:25 AM
misc-data		7/22/2026 11:58:49 AM
modules		7/17/2026 6:14:25 AM
output-20260722151947		7/22/2026 12:19:44 PM
subworkflows		7/17/2026 6:14:25 AM
test		7/17/2026 6:14:25 AM
work		7/22/2026 12:19:45 PM
workflows		7/17/2026 6:14:25 AM
basset.37797862.err	1 KB	7/22/2026 12:19:04 PM
basset.37797862.out	266 KB	7/22/2026 12:19:47 PM
bassetrun.sh	1 KB	7/21/2026 2:05:38 PM
CHANGELOG.md	1 KB	7/17/2026 6:14:24 AM
CITATION.cff	1 KB	7/17/2026 6:14:24 AM
CITATIONS.md	12 KB	7/17/2026 6:14:24 AM
CODE_OF_CONDUCT.md	3 KB	7/17/2026 6:14:24 AM
CONTRIBUTING.md	5 KB	7/17/2026 6:14:24 AM
LICENSE	2 KB	7/17/2026 6:14:24 AM
main.nf	2 KB	7/17/2026 6:14:25 AM
modules.json	3 KB	7/17/2026 6:14:25 AM
nextflow.config	10 KB	7/17/2026 6:14:25 AM
nextflow_schema.json	19 KB	7/17/2026 6:14:25 AM
pyproject.toml	1 KB	7/17/2026 6:14:25 AM
README.md	21 KB	7/17/2026 6:14:24 AM

```
cd /blue/bphl-<state>/<user>/repos/bphl-molecular/  
git clone https://github.com/BPHL-Molecular/basset  
mkdir analysis/  
cd analysis/  
cp /blue/bphl-<state>/<user>/repos/bphl-molecular/basset/*
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/

Name	Size	Changed
..		7/22/2026 12:19:47 PM
GCF_000005845.2		7/22/2026 11:57:16 AM
GCF_000006745.1		7/22/2026 11:57:40 AM
GCF_000006765.1		7/22/2026 11:57:25 AM
GCF_000006785.2		7/22/2026 11:57:36 AM
GCF_000006845.1		7/22/2026 11:57:28 AM
GCF_000006885.1		7/22/2026 11:57:34 AM
GCF_000006945.2		7/22/2026 11:57:17 AM
GCF_000008485.1		7/22/2026 11:57:41 AM
GCF_000008805.1		7/22/2026 11:57:28 AM
GCF_000009645.1		7/22/2026 11:57:21 AM
GCF_000027305.1		7/22/2026 11:57:32 AM
GCF_000196035.1		7/22/2026 11:57:21 AM
GCF_000196095.1		7/22/2026 11:57:41 AM
GCF_000240185.1		7/22/2026 11:57:24 AM
GCF_000272065.1		7/22/2026 11:57:37 AM
GCF_000285655.3		7/22/2026 11:57:18 AM
GCF_000746645.1		7/22/2026 11:57:31 AM
samplesheet.csv	3 KB	7/22/2026 12:18:15 PM

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/

Name	Size	Changed
..		7/22/2026 12:19:47 PM
GCF_000005845.2		7/22/2026 11:57:16 AM
GCF_000006745.1		7/22/2026 11:57:40 AM
GCF_000006765.1		7/22/2026 11:57:25 AM
GCF_000006785.2		7/22/2026 11:57:36 AM
GCF_000006845.1		7/22/2026 11:57:28 AM
GCF_000006885.1		7/22/2026 11:57:34 AM
GCF_000006945.2		7/22/2026 11:57:17 AM
GCF_000008485.1		7/22/2026 11:57:41 AM
GCF_000008805.1		7/22/2026 11:57:28 AM
GCF_000009645.1		7/22/2026 11:57:21 AM
GCF_000027305.1		7/22/2026 11:57:32 AM
GCF_000196035.1		7/22/2026 11:57:21 AM
GCF_000196095.1		7/22/2026 11:57:41 AM
GCF_000240185.1		7/22/2026 11:57:24 AM
GCF_000272065.1		7/22/2026 11:57:37 AM
GCF_000285655.3		7/22/2026 11:57:18 AM
GCF_000746645.1		7/22/2026 11:57:31 AM
samplesheet.csv	3 KB	7/22/2026 12:18:15 PM

sample,fastq_1,fastq_2,assembly,organism
GCF_000005845.2,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000005845.2/GCF_000005845.2.fasta,Escherichia_coli
GCF_000006945.2,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000006945.2/GCF_000006945.2.fasta,Salmonella
GCF_000285655.3,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000285655.3/GCF_000285655.3.fasta,Shigella
GCF_000009645.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000009645.1/GCF_000009645.1.fasta,Staphylococcus_aureus
GCF_000196035.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000196035.1/GCF_000196035.1.fasta,Listeria_monocytogenes
GCF_000240185.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000240185.1/GCF_000240185.1.fasta,Klebsiella_pneumoniae_complex
GCF_000006765.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000006765.1/GCF_000006765.1.fasta,Pseudomonas_aeruginosa
GCF_000006845.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000006845.1/GCF_000006845.1.fasta,Neisseria_gonorrhoeae
GCF_000746645.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000746645.1/GCF_000746645.1.fasta,Acinetobacter_baumannii
GCF_000027305.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000027305.1/GCF_000027305.1.fasta,Haemophilus_influenzae
GCF_000006885.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000006885.1/GCF_000006885.1.fasta,Streptococcus_pneumoniae
GCF_000006785.2,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000006785.2/GCF_000006785.2.fasta,Streptococcus_pyogenes
GCF_000272065.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000272065.1/GCF_000272065.1.fasta,Streptococcus_dysgalactiae
GCF_000006745.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000006745.1/GCF_000006745.1.fasta,Vibrio_cholerae
GCF_000196095.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000196095.1/GCF_000196095.1.fasta,Vibrio_parahaemolyticus
GCF_000008485.1,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/GCF_000008485.1/GCF_000008485.1.fasta,Legionella_pneumophila

Application Cont.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/

Name	Size	Changed
..		7/21/2026 12:04:46 PM
assets		7/17/2026 6:14:25 AM
bin		7/17/2026 6:14:25 AM
cache		7/22/2026 12:19:45 PM
conf		7/17/2026 6:14:25 AM
docs		7/17/2026 6:14:25 AM
input		7/22/2026 12:04:39 PM
lib		7/17/2026 6:14:25 AM
misc-data		7/22/2026 11:58:49 AM
modules		7/17/2026 6:14:25 AM
output-20260722151947		7/22/2026 12:19:44 PM
subworkflows		7/17/2026 6:14:25 AM
test		7/17/2026 6:14:25 AM
work		7/22/2026 12:19:45 PM
workflows		7/17/2026 6:14:25 AM
basset.37797862.err	1 KB	7/22/2026 12:19:04 PM
basset.37797862.out	266 KB	7/22/2026 12:19:47 PM
bassetrun.sh	1 KB	7/21/2026 2:05:38 PM
CHANGELOG.md	1 KB	7/17/2026 6:14:24 AM
CITATION.cff	1 KB	7/17/2026 6:14:24 AM
CITATIONS.md	12 KB	7/17/2026 6:14:24 AM
CODE_OF_CONDUCT.md	3 KB	7/17/2026 6:14:24 AM
CONTRIBUTING.md	5 KB	7/17/2026 6:14:24 AM
LICENSE	2 KB	7/17/2026 6:14:24 AM
main.nf	2 KB	7/17/2026 6:14:25 AM
modules.json	3 KB	7/17/2026 6:14:25 AM
nextflow.config	10 KB	7/17/2026 6:14:25 AM
nextflow_schema.json	19 KB	7/17/2026 6:14:25 AM
pyproject.toml	1 KB	7/17/2026 6:14:25 AM
README.md	21 KB	7/17/2026 6:14:24 AM

```
#!/usr/bin/bash
```

```
#SBATCH --account=bphl-umbrella
```

```
#SBATCH --qos=bphl-umbrella
```

```
#SBATCH --job-name=basset
```

```
#SBATCH --ntasks=1
```

```
#SBATCH --cpus-per-task=40
```

```
#SBATCH --mem=200gb
```

```
#SBATCH --time=48:00:00
```

```
#SBATCH --output=basset.%j.out
```

```
#SBATCH --error=basset.%j.err
```

```
#SBATCH --mail-user=Nikhil.Yengala@flhealth.gov
```

```
#SBATCH --mail-type=FAIL,END
```

```
module load java
```

```
module load conda
```

```
module load apptainer
```

```
module load nextflow/25.10.4
```

```
export NXF_APPTAINER_CACHEDIR=/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/cache/
```

```
cd /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824
```

```
nextflow run main.nf \
```

```
--input /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/samplesheet.csv \
```

```
--outdir output/ \
```

```
-profile apptainer
```

```
dt=$(date "+%Y%m%d%H%M%S")
```

```
mv output "output-${dt}"
```

Application Cont.

Path: /.../repos/bphl-molecular/analysis/basset0824/output-20260722151947/

Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

Path: /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/

Name	Size	Changed
..		7/21/2026 12:04:46 PM
assets		7/17/2026 6:14:25 AM
bin		7/17/2026 6:14:25 AM
cache		7/22/2026 12:19:45 PM
conf		7/17/2026 6:14:25 AM
docs		7/17/2026 6:14:25 AM
input		7/22/2026 12:04:39 PM
lib		7/17/2026 6:14:25 AM
misc-data		7/22/2026 11:58:49 AM
modules		7/17/2026 6:14:25 AM
output-20260722151947		7/22/2026 12:19:44 PM
subworkflows		7/17/2026 6:14:25 AM
test		7/17/2026 6:14:25 AM
work		7/22/2026 12:19:45 PM
workflows		7/17/2026 6:14:25 AM
basset.37797862.err	1 KB	7/22/2026 12:19:04 PM
basset.37797862.out	266 KB	7/22/2026 12:19:47 PM
bassetrun.sh	1 KB	7/21/2026 2:05:38 PM
CHANGELOG.md	1 KB	7/17/2026 6:14:24 AM
CITATION.cff	1 KB	7/17/2026 6:14:24 AM
CITATIONS.md	12 KB	7/17/2026 6:14:24 AM
CODE_OF_CONDUCT.md	3 KB	7/17/2026 6:14:24 AM
CONTRIBUTING.md	5 KB	7/17/2026 6:14:24 AM
LICENSE	2 KB	7/17/2026 6:14:24 AM
main.nf	2 KB	7/17/2026 6:14:25 AM
modules.json	3 KB	7/17/2026 6:14:25 AM
nextflow.config	10 KB	7/17/2026 6:14:25 AM
nextflow_schema.json	19 KB	7/17/2026 6:14:25 AM
pyproject.toml	1 KB	7/17/2026 6:14:25 AM
README.md	21 KB	7/17/2026 6:14:24 AM

sbatch bassetrun.sh

Application Cont.

../bphl-molecular/analysis/basset0824/output-20260722151947/pipeline_info/

Name	Size	Changed
..		7/22/2026 12:19:44 PM
execution_report_2026-07-22_15...	2,979 KB	7/22/2026 12:19:47 PM
execution_timeline_2026-07-22_...	253 KB	7/22/2026 12:19:47 PM
execution_trace_2026-07-22_15-...	4 KB	7/22/2026 12:19:46 PM
pipeline_dag_2026-07-22_15-19...	6 KB	7/22/2026 12:19:46 PM
samplesheet.valid.csv	3 KB	7/22/2026 12:19:13 PM
software_versions.yml	1 KB	7/22/2026 12:19:46 PM

../repos/bphl-molecular/analysis/basset0824/output-20260722151947/

Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

Nextflow workflow report

[nauseous_mendel]

Workflow execution completed successfully. One or more tasks reported an error condition.

Run times

22-Jul-2026 15:19:07 - 22-Jul-2026 15:19:46 (duration: **38.7s**)

23 succeeded

1 ign...

Nextflow command

```
nextflow run main.nf --input /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/input/samplesheet.csv --outdir output/ -profile apptainer
```

CPU-Hours

0.2 (4.6% failed)

Launch directory

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824

Work directory

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824/work

Project directory

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/basset0824

Script name

main.nf

Script ID

006e43319ba5a782de8e5f64cc0f41de

Workflow session

31b6952d-e802-4152-9fcc-a0508997fdb4

Workflow profile

apptainer

Nextflow version

version 25.10.4, build 11173 (10-02-2026 15:17 UTC)

Application Cont.

sample	organism	tool	result_type	result_value	tool_version	basset_version	analysis_date
GCF_000746645.1	Acinetobacter_baumannii	kaptive	k_antigen	K9	3.2.1	1.0.0	7/22/2026
GCF_000746645.1	Acinetobacter_baumannii	kaptive	k_locus	KL9	3.2.1	1.0.0	7/22/2026
GCF_000746645.1	Acinetobacter_baumannii	kaptive	oc_antigen	-	3.2.1	1.0.0	7/22/2026
GCF_000746645.1	Acinetobacter_baumannii	kaptive	oc_locus	OCL10	3.2.1	1.0.0	7/22/2026
GCF_000005845.2	Escherichia_coli	ectyper	h_antigen	H48	2.0.0	1.0.0	7/22/2026
GCF_000005845.2	Escherichia_coli	ectyper	o_antigen	O16	2.0.0	1.0.0	7/22/2026
GCF_000005845.2	Escherichia_coli	ectyper	serotype	O16:H48	2.0.0	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	k_antigen	ybt 9; ICEkp3	3.2.4	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	k_locus		15 3.2.4	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	o_antigen		10 3.2.4	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	o_locus		19 3.2.4	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	serogroup		1 1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	subspecies	pneumophila	1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	wzm_serogroup		1 1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	wzt_serogroup		1 1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	eL_gato	sbt		36 1.22.0	1.0.0	7/22/2026
GCF_000196035.1	Listeria_monocytogenes	lissero	serotype	1/2c, 3c	0.4.10	1.0.0	7/22/2026
GCF_000006845.1	Neisseria_gonorrhoeae	ngmaster	ng-mast/ng-star	773/-	2.0.0	1.0.0	7/22/2026
GCF_000006765.1	Pseudomonas_aeruginosa	pasty	serotype	O5	2.2.1	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	seqsero2	antigenic_profile	4:i:1,2	1.3.2	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	seqsero2	identification	Salmonella enterica subspecies enterica (subspecies I)	1.3.2	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	seqsero2	serotype	Typhimurium	1.3.2	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	antigenic_profile	1,4,[5],12:i:1,2	1.1.3	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	qc_status	PASS	1.1.3	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	serotype	Typhimurium	1.1.3	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	subspecies	enterica	1.1.3	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	cluster	Not Shigella/EIEC	1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	h_antigen		1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	ipaH	-	1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	o_antigen		1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	serotype	Not Shigella/EIEC	1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	virulence_plasmid		0 1.3.5	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	sccmec	mecA	+	1.2.0	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	sccmec	sccmecsubtype	IIa	1.2.0	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	sccmec	sccmctype	II	1.2.0	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	spaTyper	spatype	t002	0.2.1	1.0.0	7/22/2026
GCF_000272065.1	Streptococcus_dysgalactiae	emmtyper	emm_cluster	-	0.2.0	1.0.0	7/22/2026
GCF_000272065.1	Streptococcus_dysgalactiae	emmtyper	emm_type	-	0.2.0	1.0.0	7/22/2026
GCF_000006885.1	Streptococcus_pneumoniae	pbptyper	pbptype		0:00:03 2.0.0	1.0.0	7/22/2026
GCF_000006785.2	Streptococcus_pyogenes	emmtyper	emm_cluster	A-C3	0.2.0	1.0.0	7/22/2026
GCF_000006785.2	Streptococcus_pyogenes	emmtyper	emm_type	EMM1.6	0.2.0	1.0.0	7/22/2026
GCF_000006745.1	Vibrio_cholerae	kaptive	o_antigen	O1	3.2.1	1.0.0	7/22/2026
GCF_000006745.1	Vibrio_cholerae	kaptive	o_locus	OL1-Ogawa	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	k_antigen	unknown (KL6)	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	k_locus	KL6	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	o_antigen	-	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	o_locus	OL3-or-OL13	3.2.1	1.0.0	7/22/2026

/.../repos/bphl-molecular/analysis/basset0824/output-20260722151947/		
Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

Application Cont.

sample	organism	tool	result_type	result_value	tool_version	basset_version	analysis_date
GCF_000746645.1	Acinetobacter_baumannii	kaptive	k_antigen	K9	3.2.1	1.0.0	7/22/2026
GCF_000746645.1	Acinetobacter_baumannii	kaptive	k_locus	KL9	3.2.1	1.0.0	7/22/2026
GCF_000746645.1	Acinetobacter_baumannii	kaptive	oc_antigen	-	3.2.1	1.0.0	7/22/2026
GCF_000746645.1	Acinetobacter_baumannii	kaptive	oc_locus	OCL10	3.2.1	1.0.0	7/22/2026
GCF_000005845.2	Escherichia_coli	ectyper	h_antigen	H48	2.0.0	1.0.0	7/22/2026
GCF_000005845.2	Escherichia_coli	ectyper	o_antigen	O16	2.0.0	1.0.0	7/22/2026
GCF_000005845.2	Escherichia_coli	ectyper	serotype	O16:H48	2.0.0	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	k_antigen	ybt 9; ICEKp3	3.2.4	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	k_locus		15 3.2.4	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	o_antigen		10 3.2.4	1.0.0	7/22/2026
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	o_locus		19 3.2.4	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	serogroup		1 1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	subspecies	pneumophila	1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	wzm_serogroup		1 1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	abricate	wzt_serogroup		1 1.4.0	1.0.0	7/22/2026
GCF_000008485.1	Legionella_pneumophila	el_gato	sbt		36 1.22.0	1.0.0	7/22/2026
GCF_000196035.1	Listeria_monocytogenes	lissero	serotype	1/2c, 3c	0.4.10	1.0.0	7/22/2026
GCF_000006845.1	Neisseria_gonorrhoeae	ngmaster	ng-mast/ng-star	773/-	2.0.0	1.0.0	7/22/2026
GCF_000006765.1	Pseudomonas_aeruginosa	pasty	serotype	O5	2.2.1	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	seqsero2	antigenic_profile	4:i:1,2	1.3.2	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	seqsero2	identification	Salmonella enterica subspecies enterica (subspecies I)	1.3.2	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	seqsero2	serotype	Typhimurium	1.3.2	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	antigenic_profile	1,4,[5],12:i:1,2	1.1.3	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	qc_status	PASS	1.1.3	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	serotype	Typhimurium	1.1.3	1.0.0	7/22/2026
GCF_000006945.2	Salmonella	sistr	subspecies	enterica	1.1.3	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	cluster	Not Shigella/EIEC	1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	h_antigen		1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	ipaH	-	1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	o_antigen		1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	serotype	Not Shigella/EIEC	1.3.5	1.0.0	7/22/2026
GCF_000285655.3	Shigella	shigeifinder	virulence_plasmid		0 1.3.5	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	sccmec	mecA	+	1.2.0	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	sccmec	sccmecsubtype	IIa	1.2.0	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	sccmec	sccmecsubtype	II	1.2.0	1.0.0	7/22/2026
GCF_000009645.1	Staphylococcus_aureus	spaTyper	spatype	t002	0.2.1	1.0.0	7/22/2026
GCF_000272065.1	Streptococcus_dysgalactiae	emmtyper	emm_cluster	-	0.2.0	1.0.0	7/22/2026
GCF_000272065.1	Streptococcus_dysgalactiae	emmtyper	emm_type	-	0.2.0	1.0.0	7/22/2026
GCF_000006885.1	Streptococcus_pneumoniae	pbptyper	pbptype		0:00:03 2.0.0	1.0.0	7/22/2026
GCF_000006785.2	Streptococcus_pyogenes	emmtyper	emm_cluster	A-C3	0.2.0	1.0.0	7/22/2026
GCF_000006785.2	Streptococcus_pyogenes	emmtyper	emm_type	EMM1.6	0.2.0	1.0.0	7/22/2026
GCF_000006745.1	Vibrio_cholerae	kaptive	o_antigen	O1	3.2.1	1.0.0	7/22/2026
GCF_000006745.1	Vibrio_cholerae	kaptive	o_locus	OL1-Ogawa	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	k_antigen	unknown (KL6)	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	k_locus	KL6	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	o_antigen	-	3.2.1	1.0.0	7/22/2026
GCF_000196095.1	Vibrio_parahaemolyticus	kaptive	o_locus	OL3-or-OL13	3.2.1	1.0.0	7/22/2026

/.../repos/bphl-molecular/analysis/basset0824/output-20260722151947/		
Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

Application Cont.

../analysis/basset0824/output-20260722151947/Acinetobacter_baumannii/

Name	Size	Changed
..		7/22/2026 12:19:44 PM
GCF_000746645.1_kaptive_k.tsv	2 KB	7/22/2026 12:19:17 PM
GCF_000746645.1_kaptive_oc.tsv	1 KB	7/22/2026 12:19:17 PM
GCF_000746645.1_kaptive_sum...	1 KB	7/22/2026 12:19:17 PM

../repos/bphl-molecular/analysis/basset0824/output-20260722151947/

Name	Size	Changed
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

GCF_000746645.1	Acinetobacter_baumannii	kaptive	k_locus	KL9
GCF_000746645.1	Acinetobacter_baumannii	kaptive	k_antigen	K9
GCF_000746645.1	Acinetobacter_baumannii	kaptive	oc_locus	OCL10
GCF_000746645.1	Acinetobacter_baumannii	kaptive	oc_antigen	-

Application Cont.

File Explorer view of `/.../bphl-molecular/analysis/basset0824/output-20260722151947/Escherichia_coli/`

Name	Size	Changed
..		7/22/2026 12:19:44 PM
GCF_000005845.2_ectyper.log	5 KB	7/22/2026 12:19:36 PM
GCF_000005845.2_ectyper_blast...	15 KB	7/22/2026 12:19:36 PM
GCF_000005845.2_ectyper_outp...	1 KB	7/22/2026 12:19:36 PM
GCF_000005845.2_ectyper_sum...	1 KB	7/22/2026 12:19:36 PM

File Explorer view of `/.../repos/bphl-molecular/analysis/basset0824/output-20260722151947/`

Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

Summary table for `GCF_000005845.2`

GCF_000005845.2	Escherichia_coli	ectyper	o_antigen	O16
GCF_000005845.2	Escherichia_coli	ectyper	h_antigen	H48
GCF_000005845.2	Escherichia_coli	ectyper	serotype	O16:H48

Main analysis results table

Name	Species	SpeciesMashRatio	SpeciesMashDistance	SpeciesMashTopID	O-type	H-type	Serotype	QC	Evidence	GeneScores	AlleleKeys	GeneIdentities(%)	GeneCoverages(%)	GeneContigNames	GeneRanges	GeneLengths	DatabaseVer	Warnings
GCF_000005845.2	Escherichia coli	1000/1000	0	GCF_000005845.2	O16	H48	O16:H48	-	Based on 3 allele(s)	wzx:1;wzy:1;fliC:1; O16-1-wzx-origin;O16-2-wzy-origin;H48-1-fliC-origin;		100;100;100;	100;100;100;	NC_000913.3;NC_000913.3;NC_000913.3;	2108337-2109584;2106060-2107226;2002110-2003606;	1248;1167;1497;	v1.0 (11-03-2020)	-

Application Cont.

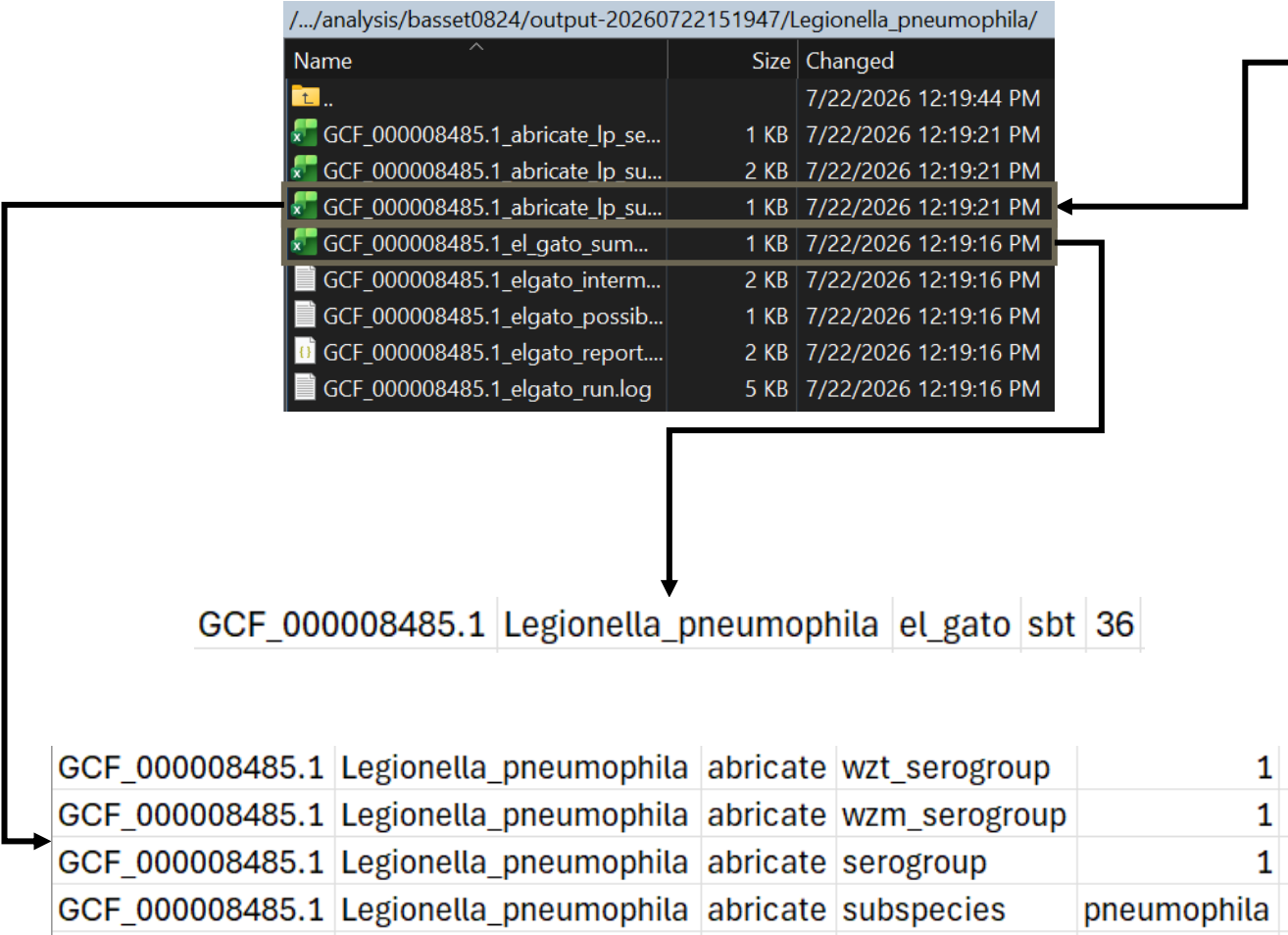
Name	Size	Changed
..		7/22/2026 12:19:44 PM
GCF_000240185.1_kleborate_out...	5 KB	7/22/2026 12:19:42 PM
GCF_000240185.1_kleborate_su...	1 KB	7/22/2026 12:19:42 PM

GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	k_locus	15
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	k_antigen	ybt 9; ICEKp3
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	o_locus	19
GCF_000240185.1	Klebsiella_pneumoniae_complex	kleborate	o_antigen	10

/.../repos/bphl-molecular/analysis/basset0824/output-20260722151947/		
Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

strain	enterobac	enterobac	general_c	general_c	general_c	general_c	general_c	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella	klebsiella
GCF_0002 Klebsiella strong			7	5333942	5333942	5682322	yes (1)	ambiguous:ST11		3	3	1	1	1	1	4	15 ybt 9; ICEK	14	11	14	5	9	22	19	10	5	11	11 -		0 -	-	-	-	-	-	-

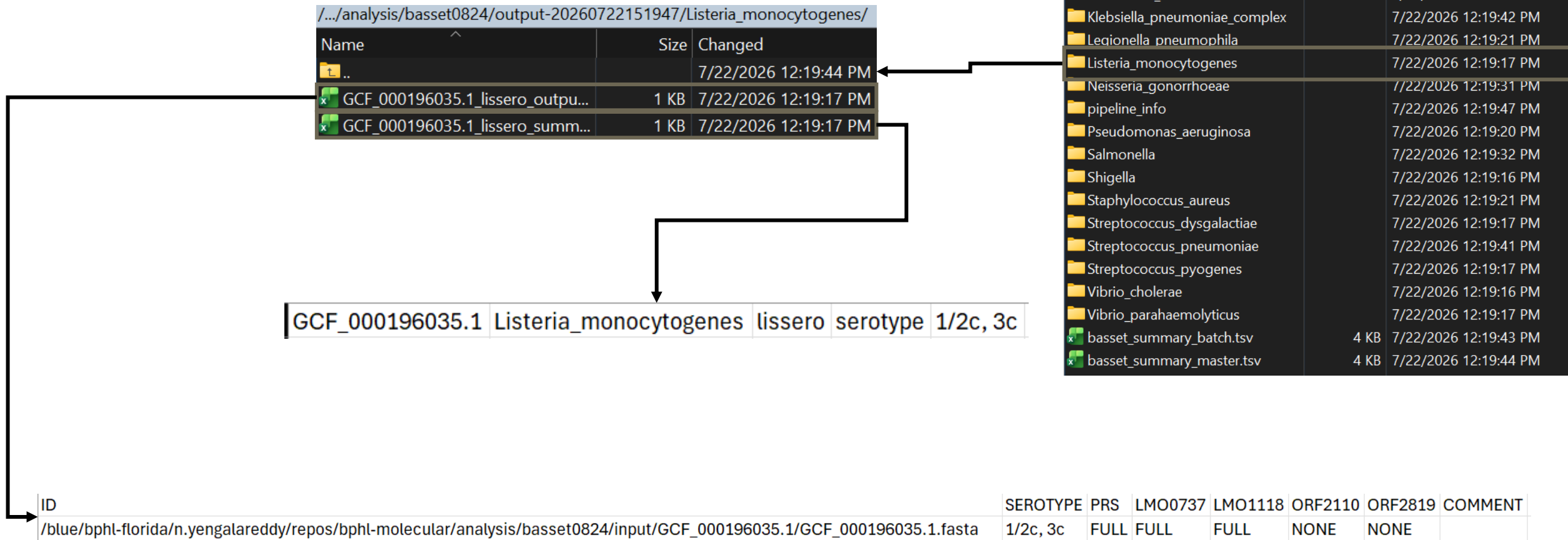
Application Cont.



Directory listing of files generated by Basset for Legionella pneumophila analysis:

Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

Application Cont.



Application Cont.

../analysis/basset0824/output-20260722151947/Neisseria_gonorrhoeae/

Name	Size	Changed
..		7/22/2026 12:19:44 PM
x GCF_000006845.1_ngmaster_out...	1 KB	7/22/2026 12:19:31 PM
x GCF_000006845.1_ngmaster_su...	1 KB	7/22/2026 12:19:31 PM

../repos/bphl-molecular/analysis/basset0824/output-20260722151947/

Name	Size	Changed
..		7/22/2026 12:19:47 PM
Acinetobacter_baumannii		7/22/2026 12:19:17 PM
Escherichia_coli		7/22/2026 12:19:36 PM
Klebsiella_pneumoniae_complex		7/22/2026 12:19:42 PM
Legionella_pneumophila		7/22/2026 12:19:21 PM
Listeria_monocytogenes		7/22/2026 12:19:17 PM
Neisseria_gonorrhoeae		7/22/2026 12:19:31 PM
pipeline_info		7/22/2026 12:19:47 PM
Pseudomonas_aeruginosa		7/22/2026 12:19:20 PM
Salmonella		7/22/2026 12:19:32 PM
Shigella		7/22/2026 12:19:16 PM
Staphylococcus_aureus		7/22/2026 12:19:21 PM
Streptococcus_dysgalactiae		7/22/2026 12:19:17 PM
Streptococcus_pneumoniae		7/22/2026 12:19:41 PM
Streptococcus_pyogenes		7/22/2026 12:19:17 PM
Vibrio_cholerae		7/22/2026 12:19:16 PM
Vibrio_parahaemolyticus		7/22/2026 12:19:17 PM
x basset_summary_batch.tsv	4 KB	7/22/2026 12:19:43 PM
x basset_summary_master.tsv	4 KB	7/22/2026 12:19:44 PM

GCF_000006845.1	Neisseria_gonorrhoeae	ngmaster	ng-mast/ng-star	773/-
-----------------	-----------------------	----------	-----------------	-------

FILE	SCHEME	NG-MAST/	porB_NG-M	mtbpB	penA	mtrR	porB_NG-S	ponA	gyrA	parC	23S	CC
GCF_0000	ngmaSTar	773/-	514	219	19	5	100	100	100	100	100,34	-

Conclusion



Fundamentals of
BaSSeT



Installation and setup
of BaSSeT in HPG



Successfully executed
job query for BaSSeT



Generated output files



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

bphl-sebioinformatics@flhealth.gov

Molly Mitchell, PhD

Bioinformatics Supervisor

Molly.Mitchell@flhealth.gov

Nikhil Reddy, MS

Bioinformatician

Nikhil.Yengala@flhealth.gov

Sam Bernhoft, MPH

Bioinformatician

Samantha.bernhoft@flhealth.gov

Arnold RodriguezHilario, MS

Bioinformatician

Arnold.RodriguezHilario@flhealth.gov